

Technical Document #103: Cybersecurity - Comprehensive Overview

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Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Vulnerability management identifies, classifies, and remediates security vulnerabilities. Regular scanning and patching are essential components.

Intrusion detection systems (IDS) monitor network traffic for suspicious activity. They alert administrators of potential security breaches.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Security information and event management (SIEM) systems collect and analyze security logs.
- Social engineering manipulates people into revealing confidential information.
- Zero trust architecture assumes no implicit trust and verifies every request.